Bharat D. Sharma, Ph.D.

Research Scientist (R & D Associate),
Computational Sciences & Engineering Division,
Oak Ridge National Laboratory, TN, USA
Early Career Representative, AAAS

Contact via: bharat.sharma.neu@gmail.com
or sharmabd@ornl.gov
https://sharma-bharat.github.io/
https://www.aaas.org/governance/section-w

PERSONAL INFORMATION

EDUCATION

Northeastern University — Boston, Massachusetts, USA
PhD, Interdisciplinary Engineering — Sep 2022
Dissertation: Analysis of Global Carbon Cycle Extremes, their Compound Climate Drivers, and Implications for Terrestrial Carbon Cycle — 4.0 (US GPA) or 10/10 CGPA
Advisors: Prof. Auroop R. Ganguly and Dr. Forrest M. Hoffman

Technical University of Munich — Munich, Germany
MS, Transportation Systems — Aug 2016
Thesis: Resilience in Urban Cities: An approach to study the interaction between evacuation and land use & transportation infrastructure. — 1.6 (German) or 8.8/10 CGPA

National Institute of Technology — Hamirpur, H.P., India
B. Tech, Civil Engineering — Jun 2012
Thesis: Characteristics Of Soil, Sand, Fly Ash And Ceramics Mix For Use As Subgrade Material. — 9.05/10 CGPA
Secured 2nd rank in the batch of 2008-12 Civil Engineering.

PROFESSIONAL EXPERIENCE

Oak Ridge National Laboratory (ORNL), Tennessee, USA — May 2025 – present
Research Scientist - Computational Urban Sciences

- Investigating agreement of extremes in global carbon cycle using Earth System Models and observations, and creating metrics for land model benchmarking for anomalies and extremes for various CMIP6 models.
- Leveraging the DOE's Energy Exascale Earth System Model (E3SM) Land Model (ELM) to simulate plant responses to CO_2 and Land Cover Change.
- Applied research at the intersection of computing and complex urban systems in the emerging environment of smart cities, energy infrastructures, smart grids, smarter mobility, emergency response, and urban resiliency.
- Leveraging statistical and machine learning techniques to identify, quantify, and explain clinical and process variations in healthcare of veterans, focusing on informing policy and practice improvements.
- Mentoring two PhD Candidates from University of Tennessee, Knoxville, working on climate-carbon cycle feedbacks under geoengineering scenarios and climate extremes impact on urban infrastructure.
- Mentoring a graduate student on building Large Language Model (LLMs) for Earth System Grid Federation (ESGF) Earth System Model data.

Oak Ridge National Laboratory (ORNL), Tennessee, USA — Oct 2022 – Apr 2025
Postdoctoral Research Associate - Terrestrial Ecosystem Ecology

- Modeling forest-CO_2 interactions with nutrients using vegetation demography model, FATES (Functionally Assembled Terrestrial Ecosystem Simulator) coupled with ELM-FATES to improve plant response to elevated CO_2 observed in the Free Air CO_2 Enrichment experiments.
- Created tools for preprocessing input data for Earth System Models (ESMs).
- Served as co-advisor for a University of Tennessee, Knoxville master student's thesis.
- Served as mentor for a PhD student, from University of Tennessee, Knoxville, on analysis of carbon extremes under geoengineering scenarios, providing guidance on advanced statistical methodologies and climate model interpretation.

University of Tennessee, Knoxville, USA Feb 2024 – present
Research Scientist, Advisor - ARPA-E RECOIL

- Investigated the resilience of intermodal US freight systems against climate extremes and disruptions, identifying critical nodes that could potentially diminish system functionality by 30%, proposing effective recovery strategies to enhance operational stability.
- Mentoring two Ph.D. students of UT Knoxville to learn, develop, and contribute to RECOIL (Resiliency and Emission Control through Optimizing Intermodal Logistics) project's goal of enhancing freight resilience to disruptions.

Obermeyer Planen + Beraten GmbH, Munich, Germany Mar 2015 – Aug 2016
Intern + Part-time employee (*Werkstudent*), Department of Rail Design and Engineering.

- Pre-feasibility analysis, including Environmental Impact Assessment and Cost-Benefit Analysis, for selecting rail corridors and terminal sites for the Košice-Vienna/Bratislava high-speed rail project.

Technical University of Munich, Munich, Germany Jun 2014 – Jul 2015
Graduate Research Assistant (*Werkstudent*), Department of Urban Structure & Transport Planning.

- Mobility cost estimation in the metropolitan region of Munich for the "MOR€CO", Mobility and Residential Costs, project to investigate accessibility and foster sustainable mobility by optimized poly-centric settlement.
- Produced a travel time matrix for Munich metropolitan's ("WAM",Wohnen Arbeiten Mobilität) project to analyze commuting patterns, supporting route optimization and improved mobility for 30,000+ residents.

GMR Airport Developers Limited, New Delhi, India Jul 2012 – Sep 2013
Executive Civil Engineer, Terminal 3, New Delhi International Airport

- In charge of quality control for relaying of runway 29/11 and taxiways with Larsen & Tubro Limited and helped finish the project 10% a head of schedule.
- Supervised civil engineering projects; preparation of Kaizen reports, and audits of contractors bill of quantities.

TEACHING

ENVE 691 Global Ecohydrology & Biogeochemistry, UTK Spring 2023
Taught tutorial on accessing the CMIP6 data and using Python to analyze the simulation outputs.

Teaching (Shared), CIVE 5363 Climate Science, Engineering Adaptation, & Policy, NEU Spring 2021
Taught lectures, created study material, designed and graded assignments and conducted tutorial sessions and mentored projects for 30 graduate students. Received excellent reviews.

CIVE 2260 Materials for the Built Environment, NEU Spring 2018
Graded assignments and quizzes and held office hours for answering queries of 58 students.

CIVE 2261 Lab for Materials for the Built Environment, NEU Spring 2018
Supervised field visits for surveying lab, graded lab reports and quizzes and held office hours for answering queries of 58 students.

CIVE 3464 Probability and Engineering Economy for Civil Engineering, NEU Spring 2018
Designed and taught tutorials, graded assignments and conducted tutorial sessions and held office hours for answering queries of 56 students.

SUPERVISION OF PH.D. AND MASTER THESIS

Ph.D.'s Thesis, Co-Advisor and Committee Member, University of Tennessee, Knoxville 2025 – present
Aliza Sharmin. Department of Industrial and Systems Engineering.
"Dynamic Disruption Resilience in Intermodal Transport Networks"

Master's Thesis, Co-Advisor and Committee Member, University of Tennessee, Knoxville 2024
Maedeh Rahimitouranposhti. Department of Industrial and Systems Engineering.
"Investigating Resiliency of Transportation Network Under Targeted and Potential Climate Change Disruptions"

KNOWLEDGE DISSEMINATION

BOOK CHAPTERS

Mary Warner, **Bharat Sharma**, Udit Bhatia, and Auroop Ganguly. (2019). Evaluation of Cascading Infrastructure Failures and Optimal Recovery from a Network Science Perspective. In: Ghanbarnejad F., Saha Roy R., Karimi F., Delvenne J. C., Mitra B. (eds) Dynamics On and Of Complex Networks III. DOOCN 2017. Springer Proceedings in Complexity. Springer, Cham. URL: https://doi.org/10.1007/978-3-030-14683-2_3

PUBLICATIONS

PEER-REVIEWED JOURNALS and CONFERENCE PROCEEDINGS

Aliza Sharmin, **Bharat Sharma**, Mustafa Can Camur, Olufemi A. Omitaomu, and Xueping Li (2026). Dynamic Disruption Resilience in Intermodal Transport Networks: Integrating Flow Weighting and Centrality Measures. Transportation Research Record, 0(0). https://doi.org/10.1177/03611981261437033.

Bharat Sharma, Abhilasha J. Saroj, Evan Scherrer, and Melissa R. Allen-Dumas (2026). Modeling Disruptions to Urban Metabolism using Interconnected Networks. Cities. *Manuscript under Review: JCIT-D-26-01489*. Preprint: https://doi.org/10.48550/arXiv.2604.06104.

Bharat Sharma and Jitendra Kumar (2025). Variational AutoEncoders Reveal Intensifying GPP Extremes in Continental United States based on CESM2 Simulations. 2025 IEEE International Conference on Data Mining Workshops (ICDMW), Washington, DC, USA. https://doi.org/10.1109/ICDMW69685.2025.00102.

Daniel Saedi Nia, Elias C. Massoud, **Bharat Sharma**, Jitendra Kumar, Nathaniel Collier, and Forrest M. Hoffman (2025). ESGF Assitant: A Domain-Specific Large Language Model for Navigating Earth System Data. 2025 IEEE International Conference on Data Mining Workshops (ICDMW), Washington, DC, USA. https://doi.org/10.1109/ICDMW69685.2025.00100.

Evan Scherrer, Melissa Allen-Dumas, and **Bharat Sharma** (2025). Analyzing Infrastructure Interdependencies Using Network-Of-Networks Modeling. In Proceedings of the 3rd ACM SIGSPATIAL International Workshop on Advances in Urban-AI (UrbanAI '25). Association for Computing Machinery, New York, NY, USA, 68–71. https://doi.org/10.1145/3764926.3771946

Maedah Rahimitouranposhti, **Bharat Sharma**, Mustafa Can Camur, Olufemi A. Omitaomu, and Xueping Li (2025). Investigating Resiliency of Transportation Network Under Targeted and Potential Climate Change Disruptions. Transportation Research Record, 0(0). https://doi.org/10.1177/03611981251355510

Elias C. Massoud, Nathaniel Collier, **Bharat Sharma**, and Forrest M. Hoffman (2024). Enhancing Photosynthesis Simulation Performance in ESMs with Machine Learning-Assisted Solvers. 2024 IEEE International Conference on Big Data (BigData), Washington, DC, USA, 2024, pp. 4351 - 4356. https://doi.org/10.1109/BigData62323.2024.10825207.

Bharat Sharma, Jitendra Kumar, Auroop R. Ganguly, and Forrest M. Hoffman (2023). Carbon Cycle Extremes Accelerate Weakening of the Land Carbon Sink in the Late 21st Century. Biogeosciences, 20, 1829 - 1841, doi.org/10.5194/bg-20-1829-2023. Highlighted in NewScientist and ORNL News.

Bharat Sharma, Jitendra Kumar, Nathaniel Collier, Auroop R. Ganguly, and Forrest M. Hoffman (2022). Quantifying Carbon Cycle Extremes and Attributing Their Causes Under Climate and Land Use & Land Cover Change from 1850 to 2300. Journal of Geophysical Research: Biogeosciences, 127, e2021JG006738, doi.org/10.1029/2021JG006738. Codes: https://zenodo.org/badge/latestdoi/413554760. Awarded Wiley Top Downloaded Article.

Bharat Sharma, Jitendra Kumar, Auroop R. Ganguly, and Forrest M. Hoffman (2022). Using Image Processing Techniques to Identify and Quantify Spatiotemporal Carbon Cycle Extremes. 2022 IEEE International Conference on Data Mining Workshops (ICDMW), Orlando, FL, USA, 2022, pp. 1136-1143, doi.org/10.1109/ICDMW58026.2022.00148.

Bharat Sharma, Jitendra Kumar, Auroop R. Ganguly, and Forrest M. Hoffman. Investigating Extreme Events of Losses Carbon Uptake under Various Future Global Socioeconomic Changes. *Manuscript in Preparation for Nature Climate Change.*

Bharat Sharma, A. Walker, R. Knox, C. Koven, E. Agee, R. Fisher, R. Oren, R. Norby, D. Ricciuto, X. Wei, and X. Yang. Investigating the CO_2 Response of Secondary-Succession Forests at Duke and Oak Ridge FACE Experiments Simulated with ELM-FATES-CNP. *Manuscript in Preparation for JGR Biogeosciences.*

Jitendra Kumar, Steckler, M., **Bharat Sharma**, Limber, R., Hargrove, W. W., and Forrest M. Hoffman. Drivers of Phenological Patterns and Variability in Heterogeneous Tropical Vegetation. *Manuscript in Preparation for Remote Sensing.*

SCHOLARSHIPS

Distinguished Dean's Fellowship 2016–17
College of Engineering, Northeastern University covering full tuition and stipend for PhD studies worth $ 83,576.

Scholarship for Outstanding Foreign Students 2014–16
Bavarian State Ministry of Sciences, Research and the Arts, Munich, Germany worth € 4100.

Brilliant Scholarship, HP, India 2008–12
Director, Vocational & Industrial Training, Himachal Pradesh, India

Ranked among top 2% of the students, All India Engineering Entrance Examination conducted (in Physics, Chemistry and Math) for undergraduate admissions in India 2008

RECOGNITION/AWARDS

Early-Career Representative, AAAS 2026–present
Atmospheric and Hydrospheric Sciences, American Association for the Advancement of Science (AAAS)

NewScientist Article 2024
Interviewed by NewScientist and paper highlighted in NewScientist article.

Wiley Top Downloaded Article 2023
My JGR-B paper was among the most downloaded article that published between 1-Jan and 31-Dec 2022.

Most Attended Biogeosciences Session 2023
My 2023 AOGS Biogeosciences session got recognition for most attended and best organized session.

ORNL News 2023
My 2023 Biogeosciences paper was highlighted in ORNL news Modeling Climate Extremes.

Travel Award 2022
Travel Award for attending the 2022 CESM Tutorial in Boulder, Colorado, USA.

Travel Award 2018
Travel Award for attending the 2018 US-IALE Annual Meeting in Chicago, Illinois, USA.

Travel Award 2018
Travel Award for attending the 2018 AOGS Annual Meeting in Honolulu, Hawaii, USA.

CONTRIBUTIONS TO THE INSTITUTE/POSITIONS OF RESPONSIBILITY

MENTORSHIP ROLES

Sheryl Arya 2026 – present
Graduate Intern at Oak Ridge National Laboratory,
Arizona State University, Arizona

Ruhaan Singh 2026 – present
Undergraduate Intern at Oak Ridge National Laboratory,
Stanford University, California

Evan Scherrer 2025
Undergraduate Intern at Oak Ridge National Laboratory,
Drake University, Iowa

Pragya Kandel 2021 – present
PhD Candidate,
University of Knoxville and ORNL Bredesen Center, Tennessee

Shamik Bhattacharya 2022 – 2024
Undergraduate Intern at Oak Ridge National Laboratory,
NC State University, North Carolina; Intern at ORNL, Tennessee

Russell Limber 2021 – 2022
PhD Candidate,
University of Knoxville and ORNL Bredesen Center, Tennessee

Morgan Steckler 2020 – 2022
Master Student
University of Knoxville and ORNL Bredesen Center, Tennessee

ORGANIZATION ROLES

Organizer, Climate Change Science Institute (ORNL) Journal Club Summer 2018
In-charge of scheduling and coordinating the paper presentations, and maintaining the website.

CONTRIBUTIONS OUTSIDE THE INSTITUTE

MENTORSHIP ROLES

Aliza Sharmin 2025 – present
PhD Candidate, University of Knoxville, Tennessee

Dr. Pragati Verma 2025
Project Scientist at DST Centre of Excellence for Climate Information,
Indian Institute of Technology Delhi, India

Maedeh Rahimitouranposhti 2024 – 2025
Master and PhD Student, University of Knoxville, Tennessee

Sophia Bailey 2020 – 2021
Undergraduate Student, Northeastern University (NEU) Boston, Massachusetts; Summer Project at NEU

ORGANIZATION ROLES

Convener, AGU 2026 Dec 2026
Global Environmental Change GC125069: Urban Climate, Air Quality, and Critical Networks: Feedbacks Shaping Resilient Cities.

Co-convener, AGU 2026 Dec 2026
Biogeosciences Session B069: New Approaches for Predicting Biogeochemical Cycles in Biological-Earth-Energy System Modeling.

Co-convener, AGU 2025 Dec 2025
Biogeosciences Session B066: New Mechanisms, Interactions, and Approaches for Predicting Global Biogeochemical Cycles.

Program Committee, DMESS 2025 Dec 2025
Committee: 11th Workshop on Data Mining in Earth System Science (DMESS 2025) at the IEEE International Conference on Data Mining (ICDM 2025)

Co-convener, AOGS 2025 Jul 2025
Biogeosciences Session BG05: Integrated Understanding of Global Carbon, Water, and Other Biogeochemical Cycles and Their Feedbacks.

Co-convener, AGU 2024 Dec 2024
Biogeosciences Session B106: The Global Carbon Cycle and Its Feedbacks with Anthropogenic Change.

Co-convener, AOGS 2024 Jun 2024
Biogeosciences Session BG05: Integrated Understanding of Global Carbon, Water, and Other Biogeochemical Cycles and Their Feedbacks.

Co-convener, AGU 2023 Dec 2023
Biogeosciences Session B33G: New Mechanisms, Feedbacks, and Approaches for Predicting Global Biogeochemical Cycles Under Climate Change and Intervention.

Co-convener, AOGS 2023 Aug 2023
Biogeosciences Session BG06: Integrated Understanding of Global Carbon and Other Biogeochemical Cycles and Their Feedbacks.

Program Committee, DMESS 2022 Nov 2022
Committee: 11th Workshop on Data Mining in Earth System Science (DMESS 2022) at the IEEE International Conference on Data Mining (ICDM 2022)

ACTIVELY REVIEWING FOR JOURNALS

- Nature Communications Earth & Environment
- Environmental Research Letters (Excellent reviewer)
- Journal of Geophysical Research: Biogeosciences
- Biogeosciences
- Earth Systems and Environment
- Geoscientific Model Development
- IEEE International Conference on Data Mining (ICDM), Workshop on Data Mining in Earth System Science (DMESS)
- International Workshop on Advances in Urban-AI (ACM UrbanAI)

PROGRAMMING/SOFTWARE SKILLS

- **Running ESM Land Models**: ELM, CLM, ELM–FATES, OLMT
- **Programming**: `Python`, Bash, Fortran, R, MATLAB, Octave
- **Toolkits/Software**: NCO, CDO, NCL, ILAMB, MPI, Dask, ArcGIS, VISSIM, AutoCAD
- **Machine Learning/Deep Learning Frameworks**: scikit-learn, xgboost
- **Version Control**: Git, Mercurial
- **Document/Web Preparation Software**: LaTeX, MS Office, Markdown, HTML

CERTIFICATIONS

- **Human Subjects/Health Privacy (HIPAA)**, VA/ORNL — Jun 2025
- **Introduction to Fortran**, LinkedIn Learning — Jun 2023
- **Community Earth System Model (CESM) Tutorial**, Boulder, CO — Aug 2022
- **Machine Learning by Stanford University**, Coursera — May 2022
- **Machine Learning, Data Science and Deep Learning with Python**, Udemy — Mar 2021
- **New Advances in Land Carbon Cycle Modeling**, Center for Ecosystem Science and Society, Northern Arizona University — July 2020

PROFESSIONAL AFFILIATIONS

American Geophysical Union (AGU)	2018 – Present
Asia Oceania Geosciences Society (AOGS)	2018 – Present
American Meteorological Society (AMS)	2020 – Present
American Association for the Advancement of Science (AAAS)	2025 – Present
Transportation Research Board (TRB) Individual Affiliate	2024 – Present
American Association of Geographers (AAG)	2016 – 2017
mobil.TUM - International Scientific Conference on Mobility and Transport, Germany	2015 – 2016

PRESENTATIONS

INVITED TALKS

"Assessing GPP Extremes and Their Environmental Drivers: Insights from Observations and CMIP6 Earth System Models" Technical Interchange Meeting (TIM), United Stats Air Force Weather. Mar 04, 2026

"Impact of Extremes on Transportation Networks". Oak Ridge National Laboratory Multisector Dynamics Workshop, Tennessee, USA. Sep 30, 2025

"Integrating Carbon Cycle Science and Artificial Intelligence: From Ecosystem Extremes to Societal Resilience". Department of Civil Engineering at Indian Institute of Technology, Delhi, India. May 29, 2025

"From Carbon Cycle Extremes to Forest Growth: Investigating Climate-Carbon Feedbacks Under Elevated CO_2". Centre of Atmospheric Sciences at Indian Institute of Technology, Delhi. May 28, 2025

"Quantifying the Changes in Carbon Cycle Extremes Due to Land Use Change and Attribution to Climate Drivers Through Year 2300." Reducing Uncertainties in Biogeochemical Interactions through Synthesis and Computation *url*. Feb 19, 2021

CONFERENCE PRESENTATIONS

Bharat Sharma, Forrest M. Hoffman, Jitendra Kumar, and Auroop Ganguly. December 18, 2025. "Assessing GPP Extremes and Their Environmental Drivers: Insights from Observations and CMIP6 Earth System Models". 2025 American Geophysical Union Fall Meeting, New Orleans LA, USA.

Bharat Sharma and Jitendra Kumar (2025). Variational AutoEncoders Reveal Intensifying GPP Extremes in Continental United States based on CESM2 Simulations. 2025 IEEE International Conference on Data Mining Workshops (ICDMW), Washington DC, USA.

Bharat Sharma, Ryan Knox, Charles Koven, Daniel M Ricciuto, Dr. Xinyuan Wei, Xiaojuan Yang, Richard J. Norby, Ram Oren, and Anthony P Walker.December 15, 2023. Simulating CO2 Responses in Even-Aged Secondary Forests at Duke and Oak Ridge FACE Experiments with ELM-FATES-CNP. 2024 American Geophysical Union Fall Meeting, Washington DC, USA.

Anthony P Walker, **Bharat Sharma**, and others. December 15, 2023. "Canopy Traits, Processes, and Plant Strategies Leading to Coexistence in the Demographic Land Surface Model ELM-FATES." 2024 American Geophysical Union Fall Meeting, Washington DC, USA.

Elias C. Massoud, Nathaniel Collier, **Bharat Sharma**, and Forrest M. Hoffman. December 18, 2023 "Enhancing Photosynthesis Simulation Performance in ESMs with Machine Learning-Assisted Solvers." 2024 IEEE International Conference on Big Data (BigData), Washington, DC, USA, 2024

Bharat Sharma, A. Walker, R. Knox, C. Koven, E. Agee, R. Fisher, R. Oren, R. Norby, D. Ricciuto, X. Wei, and X. Yang. December 15, 2023 "Investigating the CO_2 Response of Secondary-Succession Forests at Duke and Oak Ridge FACE Experiments Simulated with ELM-FATES-CNP". 2023 American Geophysical Union Fall Meeting, San Francisco, CA, USA.

Bharat Sharma, Forrest M. Hoffman, Jitendra Kumar, and Auroop Ganguly. August 2, 2023. "Comparative Analysis of Climate-driven Carbon Cycle Extremes Using Observations and CMIP6 Earth System Models". Annual Meeting of the Asia Oceania Geosciences Society (AOGS), 30 July−4 Aug 2023, SUNTEC, Singapore.

Bharat Sharma, A. Walker, R. Knox, C. Koven, E. Agee, R. Fisher, R. Oren, R. Norby, D. Ricciuto, X. Wei, and X. Yang. "Investigating the CO_2 Response of Secondary-Succession Forests at Duke and Oak Ridge FACE Experiments Simulated with ELM-FATES-CNP". 2023 ESS PI Meeting, 16-17 May 2023, Bethesda, MD, USA.

Bharat Sharma, A. Walker, R. Knox, C. Koven, E. Agee, R. Fisher, R. Oren, R. Norby, D. Ricciuto, X. Wei, and X. Yang. "Investigating the CO_2 Response of Secondary-Succession Forests at Duke and Oak Ridge FACE Experiments Simulated with ELM-FATES-CNP". Anthromes, CO_2, and Terrestrial Carbon, From the deep past to net-zero, 27-30 Mar 2023, Potomac, MD, USA.

Bharat Sharma, Jitendra Kumar, Nathan Collier, Auroop R. Ganguly, and Forrest M. Hoffman. January 9, 2023. "Increased Intensity of Carbon Cycle Extremes Driven by Land Use and Land Cover Change" (Poster ID 45), Land Use and Land Cover Change—Interactions with Weather and Climate. American Meteorological Society's 36th Conference on Climate Variability and Change. Denver, CO, USA.

Bharat Sharma, Jitendra Kumar, Forrest M. Hoffman, and Auroop R. Ganguly. December 15, 2022. "Quantifying Extremes in Net Biospheric Production and Attribution to Compound Climate Drivers" (B42H-1732). New Mechanisms, Feedbacks, and Approaches for Predicting Global Biogeochemical Cycles Under Climate Change and Intervention (B42H), American Geophysical Union Fall Meeting. Chicago, IL, USA.

Pragya Kandel, **Bharat Sharma**, Jitendra Kumar, and Forrest M. Hoffman. December 16, 2022. "Drought Susceptibility and Response Across Different Vegetation Types in California" (H55A-04). Evapotranspiration (ET): Advances in in Situ ET Measurements and Remote-Sensing-Based ET Estimation, Mapping, and Evaluation (H55A), American Geophysical Union Fall Meeting. Chicago, IL, USA. AGU-iposter. **Received Best Student Presentation Award.**

Shamik Bhattacharya, Forrest M. Hoffman, **Bharat Sharma**, Nathan Collier, and Min Xu. December 15, 2022. "Using Statistical Learning Methods to Accelerate Model Parameter Sensitivity Experiments" (B42H-1731). New Mechanisms, Feedbacks, and Approaches for Predicting Global Biogeochemical Cycles Under Climate Change and Intervention (B42H), American Geophysical Union Fall Meeting. Chicago, IL, USA. AGU-poster

Bharat Sharma, Jitendra Kumar, Auroop R. Ganguly, and Forrest M. Hoffman. November 28, 2022. "Using Image Processing Techniques to Identify and Quantify Spatiotemporal Carbon Cycle Extremes (S10106)." 10th Workshop on Data Mining in Earth System Science (DMESS 2022). IEEE International Conference on Data Mining Workshops (ICDMW 2022). Orlando, FL, USA. Proceedings of the 2019 IEEE International Conference on Data Mining Workshops (ICDMW 2022).

Bharat Sharma, Jitendra Kumar, Forrest M. Hoffman, and Auroop R. Ganguly. December 17, 2021. "Investigating Variability in the Intensity, Direction, and Spatial Distribution of Carbon Cycle Extremes and Attribution to Climate Drivers Using Observations and CMIP6 Earth System Models." Improving Earth System Predictability (B041), American Geophysical Union Fall Meeting. New Orleans, LA.

Morgan Steckler, **Bharat Sharma**, Forrest M. Hoffman, William W. Hargrove and Jitendra Kumar. December 14, 2021. "Effects of meteorological and ecological disturbances on tropical vegetation phenology." Understanding Phenological Responses and Feedbacks in Terrestrial Vegetation: Patterns, Mechanisms, and Consequences (B33D), American Geophysical Union Fall Meeting. New Orleans, LA.

Bharat Sharma, Jitendra Kumar, Forrest M. Hoffman, and Auroop R. Ganguly. December 12, 2020. "Detection and Attribution of Climate-Driven Extremes in Net Biome Productivity from 1850 through 2100." Abstract B019-0009 presented at American Geophysical Union (AGU) Fall Meeting (December 1–17, 2020). vspace0.3mm

Bharat Sharma, Forrest M. Hoffman, Jitendra Kumar, and Auroop R. Ganguly. December 13, 2018. "Cumulative Impacts of Human-Induced Changes on Carbon Cycle Extremes." Abstract 368411 presented at the 100th American Meteorological Society (AMS) Annual Meeting, In Robert Dickinson Symposium (January 12–16, 2020), Boston, Massachusetts, USA.

Bharat Sharma, Forrest M. Hoffman, Jitendra Kumar, Nathan Collier, and Auroop R. Ganguly. December 13, 2018. "Impact of Changes in Anthropogenic Forcing on the Terrestrial Carbon Budget through the Year 2300." Abstract B41I-2830 presented at the 2018 American Geophysical Union (AGU) Fall Meeting (December 10–14, 2018), Washington, District of Columbia, USA.

Bharat Sharma, Forrest M. Hoffman, Jitendra Kumar, Nathan Collier, and Auroop Ganguly. June 7, 2018. "Quantifying the Effect of Changes in Climate-Driven Carbon Cycle Extremes on the Terrestrial Carbon Budget through Year 2300." 15th Annual Meeting of the Asia Oceania Geosciences Society (AOGS) (June 3–8, 2018), Hawai'i Convention Center, Honolulu, Hawai'i, USA.

Bharat Sharma, Forrest M. Hoffman, Jitendra Kumar, Nathan Collier, and Auroop Ganguly. April 11, 2018. "Identification of Spatio-temporal Contiguous Carbon Cycle Extreme Events." 2018 U.S.-International Association for Landscape Ecology (US-IALE) Annual Meeting (April 8–12, 2018), Chicago, Illinois, USA.

Bharat Sharma, Forrest M. Hoffman, Jitendra Kumar, and Auroop R. Ganguly. December 15, 2017. "Carbon Cycle Extremes in the 22nd and 23rd Century and Attribution to Climate Drivers." Abstract B53J-02 presented at the 2017 American Geophysical Union (AGU) Fall Meeting (December 11–15, 2017), New Orleans, Louisiana, USA.

Bharat Sharma, Mary E. Warner, Udit Bhatia, and Auroop R. Ganguly. December 15, 2017. "Cascading Interdependencies of Built and Societal Systems." In: Symposium on Human Dynamics in Smart and Connected Communities: Spatial-Social Networks in GIS. 2017 American Association of Geographers (AAG) Annual Meeting (April 5–9, 2017), Boston, Massachusetts, USA.

CONNECT VIA

: @BharatSharmaPhD
: bharat-sharma
: 0000-0002-6698-2487
: sharma-bharat

REFERENCES

- **Dr. Forrest M. Hoffman**,
  Distinguished Computational Earth System Scientist, ORNL
  hoffmanfm@ornl.gov, +1-865-576-7680
- **Prof. Auroop R. Ganguly**,
  Distinguished Professor, Civil and Environmental Engineering, Northeastern University
  a.ganguly@northeastern.edu, +1-617-373-6005
- **Dr. Jitendra Kumar**,
  Senior R&D Staff Scientist, ORNL
  kumarj@ornl.gov, +1-865-574-9467